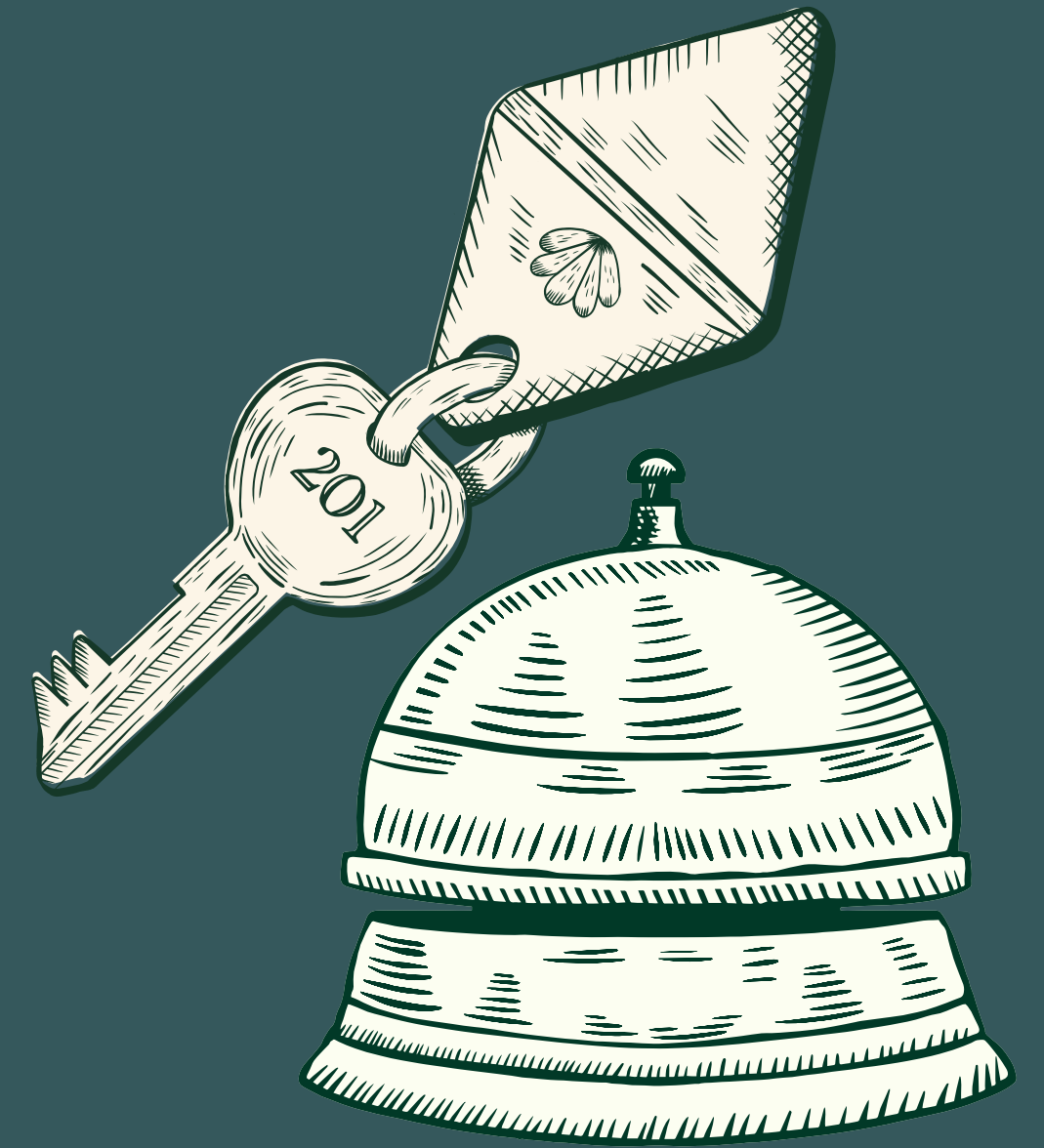


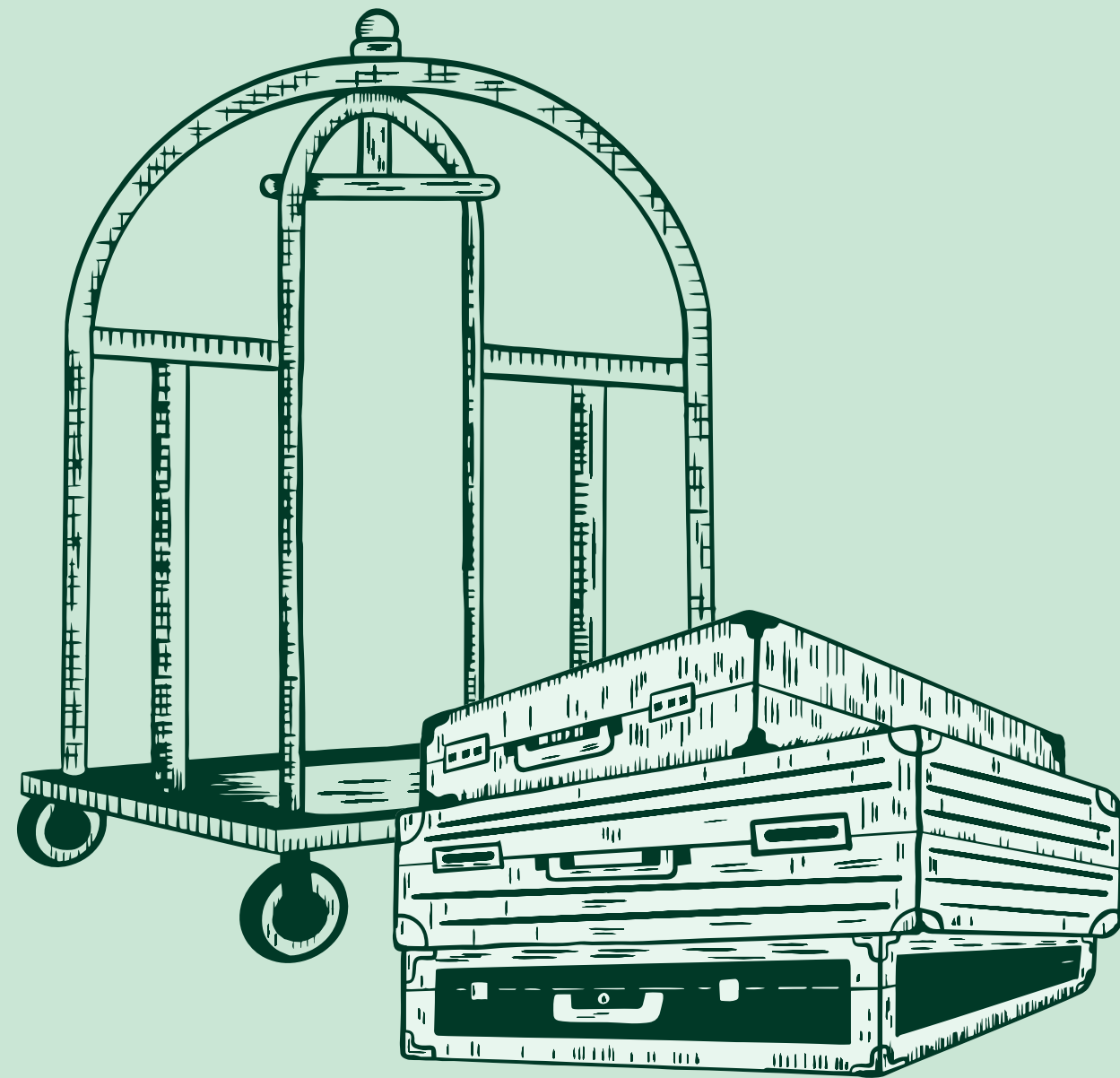
A Revenue Decision Support Tool for Hotels

BY:

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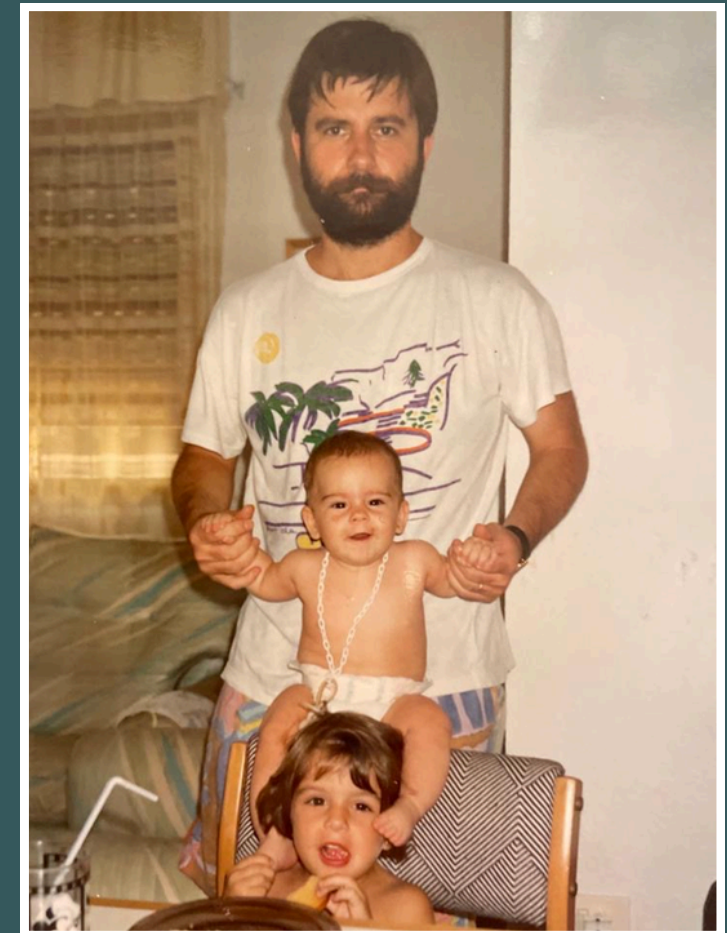
Why hotels matter to me?



1991



2002



1992



Objectives



BUSINESS PROBLEM

- Hotels lose revenue due to unpredictable booking cancellations

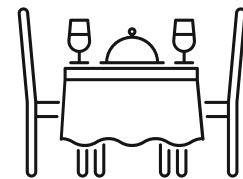
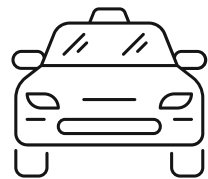


OBJECTIVE

- Develop a Decision Support Tool capable of:
- estimating cancellation probability
 - translating predictions into actionable business recommendations

Let's have a look at the numbers

Deciding the Machine Learning Model



PREPROCESSING

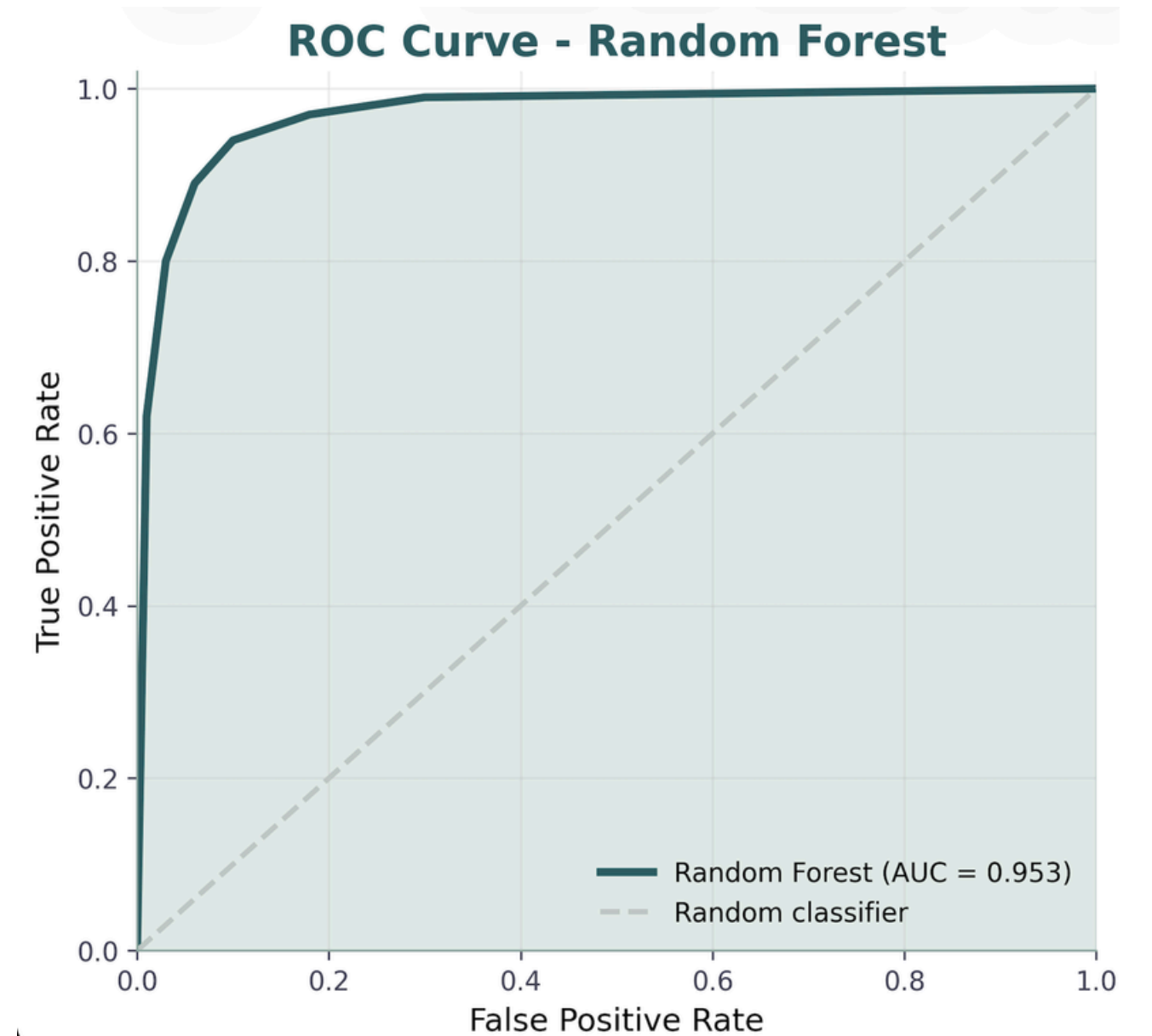
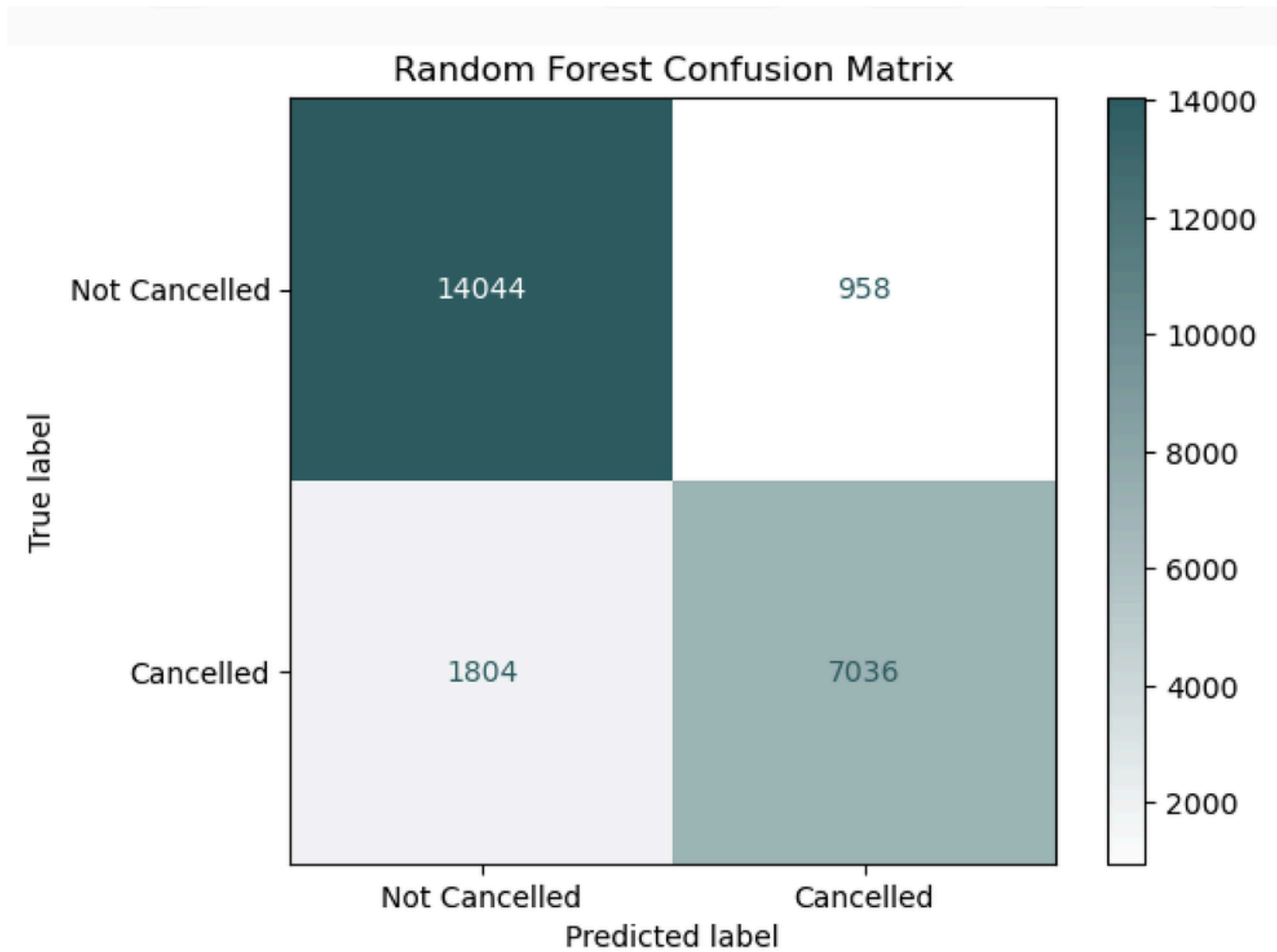
LOGISTIC REGRESSION
MODEL

RANDOM FOREST
MODEL

MODEL COMPARISON

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.811	0.809	0.640	0.715	0.888
Random Forest	0.884	0.880	0.796	0.836	0.953

Random Forest - results



The model predicts probability, but
managers need actions

Conclusions

CAN WE IDENTIFY CANCELLATION RISK BEFORE IT HAPPENS?

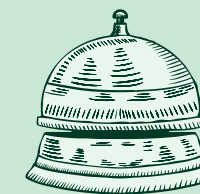


Yes



LIMITATIONS

- Historical data 2015 - 2017
- Only 2 hotels in Portugal



NEXT STEPS

- Model optimization
- Cancellation forecast for a specific period of time

Thank you!



By Elena Garrigues Pascual